Kinyarwanda News Topic Classification: Baselines, Recurrent Models and African-Language Transformers on a Deduplicated KINNEWS

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GitHub repository: https://github.com/amaliza-shal/Text-Classification-Summative
Demo video: [[VIDEO_LINK]]
Live system: [[APP_LINK]]

1. Abstract

This project classifies Kinyarwanda news articles into 14 topics using the KINNEWS dataset. Before training, we audited the data. More than half of the training rows turned out to be duplicates, 1,453 rows gave the same article two different labels, and half of the test set came from news websites that never appear in training. After cleaning, we had 6,716 training, 747 validation and 3,456 test articles. We ran eight experiments, going from simple TF-IDF baselines to a BiLSTM with Word2Vec embeddings and two small fine-tuned Transformers (AfroXLMR-mini and mMiniLM, the model it was built from). The best model was a linear SVM on character n-grams, with 76.1% accuracy and 0.577 macro-F1. Word2Vec helped the BiLSTM by 8 macro-F1 points, but it still didn't beat the SVM. The Transformers, which we could only train on a laptop CPU with short inputs, reached 67.0% accuracy and 0.390 macro-F1. The error analysis shows that the news website matters most: the SVM gets 89% right on a website it has seen before, but only 60-68% on new ones. We also found that the ~88% accuracy reported in earlier work can only be reproduced when duplicate articles leak into the evaluation.

2. Introduction

Kinyarwanda is a Bantu language and one of the official languages of Rwanda. Like most African languages, it has very little data for NLP. Joshi et al. (2020) place most of the world's languages, including most African ones, in the groups with almost no labelled data. Rwandan news websites publish Kinyarwanda articles every day, but the tools for organising text like this are mostly built for English or French.

This project is about **news topic classification**: given the title and body of a Kinyarwanda article, predict which of 14 topics it belongs to (politics, sport, economy, health, entertainment, history, technology, tourism, culture, fashion, religion, environment, education or relationship). A classifier like this could help newsrooms tag and route articles, help archives index Kinyarwanda content, and give researchers labelled data for other work. The task is also hard in ways that are typical for low-resource languages. Kinyarwanda is agglutinative, so the same stem shows up with many different prefixes and suffixes. Its spelling isn't fully standardised. And the labelled data we have is small and very unbalanced.

Objective. The goal was to build and deploy a working Kinyarwanda topic classifier and, along the way, answer four questions. (1) How reliable is the benchmark data, and how should models be evaluated on it? (2) Do word features or character n-gram features suit Kinyarwanda better? (3) Do neural models (a BiLSTM with Word2Vec, and fine-tuned Transformers) beat a well-tuned linear baseline? (4) Does a Transformer adapted to African languages do better than the same model without that adaptation?

3. Related Work

Kinyarwanda text classification. Niyongabo et al. (2020) released KINNEWS and KIRNEWS, the first news classification datasets for Kinyarwanda and Kirundi. On KINNEWS they report about 88.5% accuracy with both a TF-IDF + SVM model and a BiGRU with Word2Vec embeddings, and they also study transfer between the two languages. Other labelled Kinyarwanda data includes the sentiment set in AfriSenti (Muhammad et al., 2023). MasakhaNEWS (Adelani et al., 2023) covers news topic classification for 16 African languages, including Kirundi but not Kinyarwanda, and shows that models pretrained on African languages do well on this kind of task.

Text representations. Linear classifiers on TF-IDF n-grams are still strong baselines for topic classification. Word embeddings such as Word2Vec (Mikolov et al., 2013) give each word a dense vector learned from context. Recurrent networks such as LSTMs (Hochreiter & Schmidhuber, 1997) read text in order, and attention (Bahdanau et al., 2015; Yang et al., 2016) lets the model focus on the words that matter most. Today, pretrained Transformer encoders such as BERT (Devlin et al., 2019) and the multilingual XLM-R (Conneau et al., 2020) are the usual choice for classification.

Models for African languages. General multilingual models don't cover African languages well. AfriBERTa (Ogueji et al., 2021) was pretrained only on African languages. AfroXLMR (Alabi et al., 2022) took XLM-R-style models and continued training them on 17 African languages, including Kinyarwanda. KinyaBERT (Nzeyimana & Niyongabo Rubungo, 2022) adds a morphological analyser built specifically for Kinyarwanda. Small versions such as AfroXLMR-mini, based on the distilled mMiniLMv2 (Wang et al., 2021), make fine-tuning possible on ordinary hardware.

Existing systems and the gap. The published KINNEWS results come from classical and recurrent models trained on the dataset as released. There are also several recent community models on the Hugging Face Hub fine-tuned on KINNEWS, for example henryparfait/kinnews-topic-classifier and Seziberagabriel/kinyarwanda-news-classifier. Some of them remove duplicates too, but their model cards report a single score and don't look at where the errors come from. Duplicated data is known to inflate scores (Lee et al., 2022). This project adds four things: (i) an explanation of why published results are much higher than deduplicated ones, (ii) the finding that half of the test set comes from websites never seen in training, and what that costs each model, (iii) a matched comparison between an African-adapted Transformer and the model it started from, and (iv) an error analysis showing that many mistakes come from websites using the same category names differently.

4. Dataset and Data Preparation

4.1 Source and categories

KINNEWS (Niyongabo et al., 2020; MIT licence) is a collection of real news articles from 15 Rwandan news websites and 5 newspapers that publish in Kinyarwanda. The publishers used 48 categories between them. The dataset authors merged these into 14, and two native-speaker annotators with linguistics degrees labelled each article from its title and content, leaving out articles they weren't sure about. We used the raw version from the Hugging Face Hub, which keeps the title, body, URL and original casing, with its official split of 17,014 training and 4,254 test rows. No synthetic or machine-generated data was used anywhere in the project.

4.2 Data quality audit

Before training anything, we checked the data for duplicates, conflicting labels, encoding problems and where each article came from. What we found (Table 1) shaped every step after that.

	train	test
rows in the release	17,014	4,254
rows with < 10 words	0	4
rows whose article appears under two different labels	1,453	0
exact duplicate rows	8,098	791
test articles also in train	–	3
unique articles kept	**7,463**	**3,456**

Table 1. Effect of the cleaning steps on the official KINNEWS splits.

More than half of the training rows are repeats, and one article appears 80 times. About 70 articles also had broken characters (for example `â€™` instead of `’`), caused by UTF-8 text being read as Latin-1. The most important finding was about sources. The training articles come from 18 websites, but the test set comes from only three: muhabura.rw (46%), igihe.com (35%) and kigalipost.com (19%). **The last two never appear in training.** So most of the test set really measures how well a model copes with a news outlet it has never seen, with its own writing style and its own idea of what each category means.

4.3 Preprocessing and splits

Each article is turned into one input: `title . body`. For the TF-IDF, Word2Vec and BiLSTM models, we fix the broken characters, apply Unicode NFKC normalisation, lowercase everything, remove URLs and numbers, and replace apostrophes and punctuation with spaces. That last step matters for Kinyarwanda, where vowels are dropped and marked with an apostrophe: *n'abana* ("and the children") becomes *n abana*, so *abana* is treated as the same word wherever it appears. For the Transformers we only fix the encoding and remove URLs, because their tokenizer was trained on normal text with capitals and punctuation. We then drop articles with fewer than 10 words, every row with conflicting labels and all exact duplicates, and remove test articles that also appear in training. Finally, a stratified 10% of the training data becomes the validation set, which gives **6,716 training, 747 validation and 3,456 test** articles. All choices about epochs and checkpoints were made on the validation set; the test set was only used for the final scores.

4.4 Characteristics

The classes are very unbalanced (Figure 1). Politics has 1,590 training articles, while tourism, fashion and environment have only 30-34 each. The test set also has a different mix: for example, there are 90 fashion articles in the test set but only 34 in training. The median article is about 300 words long. The training vocabulary has 139,883 different words, and 55% of them appear only once; 5.1% of the words in the test set never appear in training. Both numbers come from Kinyarwanda's rich word formation.

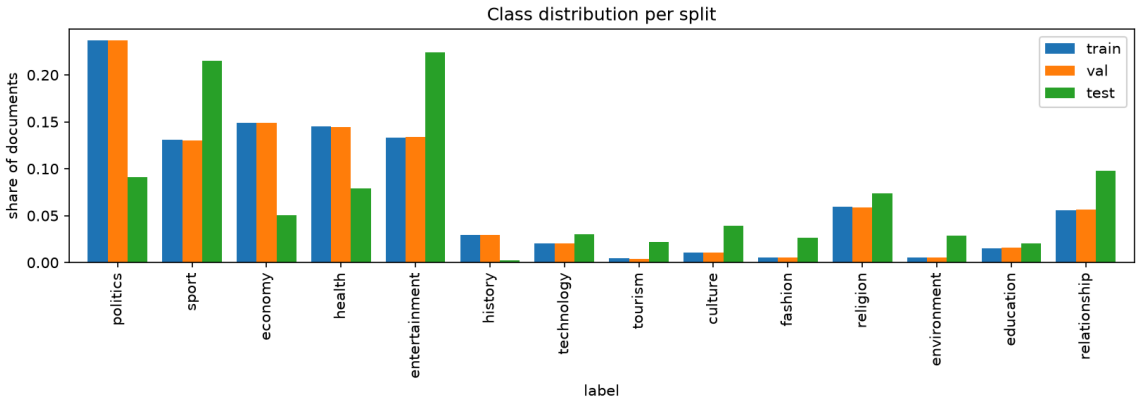


Figure 1. Share of each topic in the train, validation and test splits after deduplication.

5. Methodology

Each experiment changes one thing compared to the one before it (Table 2), so we can see what that change does. All models use the same splits, the same evaluation code and the same random seed (42).

ID	Model	Question
E0	Majority class	What is the lowest possible score?
E1	Word 1-2-gram TF-IDF + logistic regression	Baseline
E2	E1 + balanced class weights	Does reweighting help the rare topics?
E3	Character 2-5-gram TF-IDF + linear SVM (balanced)	Do sub-word features suit Kinyarwanda better?
E4	BiLSTM + attention, random embeddings	Does reading word order beat bag-of-words?
E5	BiLSTM + attention, Word2Vec embeddings	Do pretrained embeddings help with little data?
E6	AfroXLMR-mini, fine-tuned (lr 5e-5)	Does an African-language Transformer do better?
E7	mMiniLM (the model E6 was built from), lr 5e-5	Does the African-language training itself matter?
E8	E6 with learning rate 1.5e-4	Was E6 undertrained?

Table 2. Experiments and the question each one answers.

Baselines (E1-E3). All three use TF-IDF with sublinear term frequency. E1 and E2 use single words and word pairs (seen in at least 2 articles, at most 200k features) with logistic regression (C = 10); E2 also weights each class by the inverse of its size. E3 uses character 2-5-grams within words (at most 300k features) with a linear SVM (C = 0.5). Character n-grams let related forms such as *abanyeshuri*, *abanyeshuli* and *umunyeshuri* share features. We chose E1 as the baseline because it is simple, well understood and the same type of model Niyongabo et al. (2020) report.

Word2Vec. We trained skip-gram embeddings (Mikolov et al., 2013) on the training split only, so no test text was used: 100 dimensions, window 5, 10 negative samples, 10 epochs and a minimum count of 2, giving vectors for 63k words. Their nearest neighbours show both spelling variants (*kiliziya* "church" → *gatolika, gaturika, gatulika*) and related topics (*abanyeshuri* "students" → *abarimu* "teachers", *mashuri* "schools").

BiLSTM with attention (E4, E5). Each article is cut to its first 400 words (the median is about 300) and each word is mapped to an id; words seen fewer than twice become UNK. The ids go through a 100-dimensional embedding layer (random in E4, Word2Vec in E5), dropout of 0.4, and a bidirectional LSTM with 128 units in each direction. An attention layer then scores every position, and the weighted sum of the hidden states becomes a 256-dimensional summary of the article. After another dropout, a linear layer outputs a score for

each of the 14 topics. We trained with Adam (learning rate 2e-3), batch size 32, 12 epochs, gradient clipping at 1.0 and a class-weighted loss, and kept the epoch with the best validation macro-F1.

Transformers (E6-E8). AfroXLMR-mini (Alabi et al., 2022) is a 12-layer encoder with 384-dimensional hidden states. It was made by taking mMiniLMv2 (Wang et al., 2021), a small model distilled from XLM-R Large (Conneau et al., 2020), and training it further on 17 African languages, including Kinyarwanda. E7 fine-tunes the original mMiniLMv2 with exactly the same settings, so any difference between the two comes from that African-language training. The text is split into SentencePiece subwords (a 250k vocabulary), and the representation of the first token (`<s>`) is passed to a classification layer. We fine-tuned with the Hugging Face Trainer (Wolf et al., 2020): 128 tokens per article, batch size 16, 4 epochs, AdamW with 10% warm-up and linear decay, weight decay 0.01 and a class-weighted loss, keeping the best epoch on validation. All training ran on a laptop CPU (Intel Core 7 240H, 8 GB RAM). To make it fit, we froze the word-embedding layer (96M of the 118M parameters) and kept inputs to 128 tokens. With 256 tokens and trainable embeddings, a single epoch took more than two hours and ran out of memory.

Metrics. We report accuracy, precision, recall and F1 for each class, their macro averages, and weighted F1. **Macro-F1 is the main metric** because the classes are so unbalanced: it counts every topic equally, while accuracy is dominated by the five largest topics.

5.1 Existing resources and our own work

We used the following as they are: the KINNEWS dataset (Niyongabo et al., 2020); the pretrained weights of AfroXLMR-mini and mMiniLMv2; and the libraries scikit-learn (Pedregosa et al., 2011), gensim (Řehůřek & Sojka, 2010), PyTorch (Paszke et al., 2019), Hugging Face Transformers (Wolf et al., 2020) and Streamlit. Our own work is the data audit and cleaning pipeline, the train/validation/test splits, the design and code of all eight experiments (including the BiLSTM-attention model and its training loop, and the Transformer fine-tuning setup), the evaluation and error analysis, the comparison with published results, and the web application.

6. Experiments and Results

ID	Model	Accuracy	Macro P	Macro R	Macro F1	Weighted F1	Val. macro F1
E0	Majority class	0.091	0.006	0.071	0.012	0.015	0.027
E1	Word TF-IDF + LogReg	0.691	0.567	0.460	0.411	0.657	0.579
E2	E1 + class weights	0.719	0.679	0.526	0.487	0.699	0.646
E3	Char TF-IDF + LinearSVC	**0.761**	0.654	**0.610**	**0.577**	**0.753**	**0.736**
E4	BiLSTM, random emb.	0.629	0.560	0.488	0.467	0.648	0.616
E5	BiLSTM, Word2Vec emb.	0.674	0.591	0.592	0.545	0.690	0.665
E6	AfroXLMR-mini, lr 5e-5	0.670	0.405	0.449	0.390	0.634	0.445
E7	mMiniLM, lr 5e-5	stopped after 2.6 epochs: did not learn (validation loss 2.65 ≈ ln 14)					0.027
E8	AfroXLMR-mini, lr 1.5e-4	stopped after 1 epoch: collapsed to one class (validation loss 2.76)					0.008

Table 3. Test-set results (3,456 articles) and best validation macro-F1.

Reweighting (E1 → E2). Class weights raised macro-F1 from 0.41 to 0.49. Without them, the model never correctly predicted fashion or environment (F1 = 0): with only 34 examples each, these topics barely affect the loss.

Characters vs. words (E2 → E3). Switching to character n-grams gave the biggest single improvement, from 0.49 to 0.58 macro-F1 and from 0.72 to 0.76 accuracy. Since 55% of words appear only once, word features are

very sparse. Character n-grams share information between different forms of the same word and between spelling variants. This fits with the strong SVM results in Niyongabo et al. (2020).

Embeddings for the BiLSTM (E4 → E5). Starting from Word2Vec embeddings added 8 macro-F1 points, and the best validation score came earlier (epoch 7 instead of 11). With only 6.7k articles, most words are too rare for the model to learn good embeddings from the classification task alone. Both BiLSTMs overfit quickly: training loss drops to 0.02 while validation F1 levels off (Figure 2). Overall the BiLSTM still doesn't beat the character-n-gram SVM, although it does better on the four smallest topics (Section 7). A more complex model isn't automatically better when there is little data.

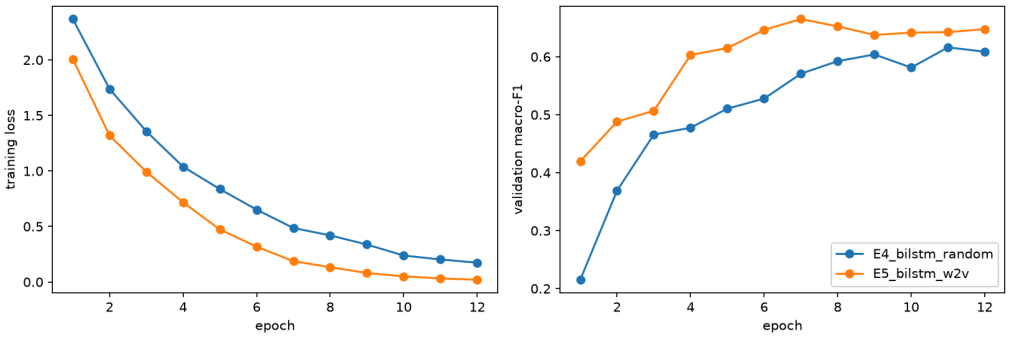


Figure 2. BiLSTM training loss and validation macro-F1 per epoch.

Transformers (E6-E8). AfroXLMR-mini at a learning rate of 5e-5 (E6) learned slowly but steadily: validation macro-F1 went 0.21, 0.32, 0.41 and 0.45 over the four epochs, and it was still improving when training stopped. On the test set it reaches 67.0% accuracy, close to the BiLSTM, but only 0.390 macro-F1. It handles the big topics well (sport 0.92, entertainment 0.79, relationship 0.78 F1), but it never predicts culture or fashion and almost never tourism or environment.

The matched control, mMiniLM without the African-language training (E7), didn't learn at all at the same learning rate. After two epochs its validation loss was still at chance level (2.65 ≈ ln 14) and its macro-F1 was 0.03. This suggests the African-language training made AfroXLMR-mini easier to fine-tune on Kinyarwanda, but we never compared the two at a setting where both work. Raising the learning rate to 1.5e-4 (E8) didn't help either: the model collapsed to predicting a single topic in the first epoch (validation macro-F1 0.008). We stopped E7 and E8 early because of time and report their validation curves instead. Under our compute limits (128 tokens, which is about 50 words, frozen embeddings and 4 epochs), the Transformers were undertrained and very sensitive to the learning rate. The TF-IDF model reads the whole article, and here it was both faster and more accurate.

6.1 Comparison with published results

Niyongabo et al. (2020) report about 88.5% accuracy, while our best model reaches 76.1%. To understand this gap, we trained a word-unigram TF-IDF SVM similar to theirs and evaluated it in three ways. On a random 9:1 split of the *raw* training data it reaches **89.2%** accuracy, but in that split 57% of the "unseen" articles also appear in the training part. On the same kind of split after removing duplicates it reaches 77.8%, and on the official test set 70-73%. So scores close to 88% only appear when duplicates leak between training and evaluation. We can't know exactly how the original numbers were calculated, but our results come from the stricter setup and shouldn't be compared directly with the published ones.

7. Error Analysis and Discussion

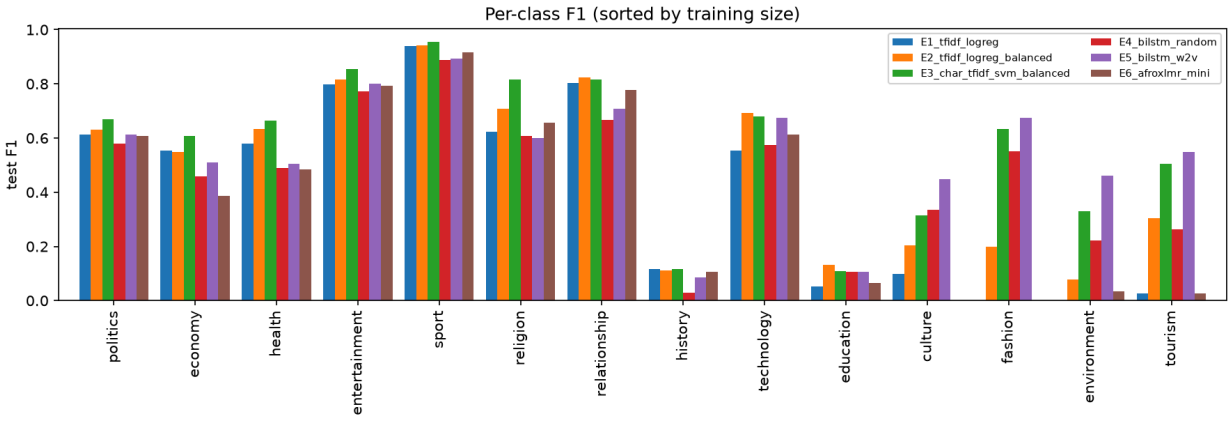


Figure 3. Test F1 per class for each model, with classes sorted by training size.

Rare topics. F1 mostly follows the amount of training data (Figure 3). Sport (0.96), entertainment (0.85), religion and relationship (0.82) are easy, while history (0.12) and education (0.11) are the hardest. The Word2Vec BiLSTM beats the SVM on the four smallest topics (culture 0.45 vs 0.31, fashion 0.67 vs 0.63, environment 0.46 vs 0.33, tourism 0.55 vs 0.50). Dense embeddings seem to help it generalise from very few examples, while the SVM wins on the common topics.

New websites are the biggest problem. The SVM gets 0.89 accuracy on muhabura.rw, a website that also appears in training, but only 0.68 on igihe.com and 0.60 on kigalipost.com, which don't. The BiLSTM does about as well on the new websites (0.67 and 0.59) but much worse on the familiar one (0.71). So part of the SVM's lead comes from wording that is specific to the websites it trained on and doesn't carry over to new outlets.

Websites define categories differently. Reading the most common confusions showed that many "errors" are really disagreements about what a category means. 13 of the 41 education → relationship errors are dream interpretations (*Ibisobanuro by'inzozi…*) or name meanings, which one test website files under education but which never appear under that label in training. Most of the 27 health → politics errors are crime and accident reports (murders, arrests by the Rwanda Investigation Bureau) that one test website labels *health*. A model that perfectly learned the training labels would still get these wrong.

Overlapping topics. The most common confusion is culture → history (48 articles). Many of these are about the Rwandan monarchy or traditional ceremonies, and some titles even start with *Amateka* ("history"). Fashion → entertainment errors are mostly celebrity stories. These topics really do overlap, and one label per article can't capture that.

Length and vocabulary. Accuracy is about 0.79 for articles of up to 300 words, but 0.68 for 500-1,000 words and 0.53 for longer ones. Long articles often cover several topics, and the neural models also cut them short. Unseen words mattered less than we expected: wrongly classified articles had only slightly more of them than correct ones (5.1% vs 4.5%).

What the model gets right. Articles with clear topic words are classified reliably, even for rare topics: football reports (*Amavubi, APR FC, umukino*), church news (*Kiliziya, Paruwasi*) and tourism rankings (*abakerarugendo*, "tourists"). The Transformer shows the website effect most strongly: 0.85 accuracy on muhabura.rw but only 0.51-0.52 on the two new websites, probably because it reads only the start of each article and learned less in its short training.

These findings match the literature. Duplicated data can inflate reported scores (Lee et al., 2022). Small, unbalanced data from many sources limits every type of model. And sub-word information matters for

languages with rich word formation, which is exactly why KinyaBERT was designed around Kinyarwanda morphology (Nzeyimana & Niyongabo Rubungo, 2022).

8. Deployment

The final system is a Streamlit web app (`webapp/streamlit_app.py`) hosted on Streamlit Community Cloud ([[APP_LINK]]). We first planned to use Hugging Face Spaces, but hosting interactive apps there now needs a paid plan, so we used Streamlit's free hosting, which the assignment allows. The trained models are stored in a free Hugging Face model repository (https://huggingface.co/amaliza-shal/kinnews-models), and the app downloads them when it starts.

The app has four pages. **Overview** explains the problem, why we chose Kinyarwanda, the 14 topics and how the system works. On **Classify**, the user pastes a Kinyarwanda article or picks one of six real test articles, and chooses either the final model (E3, the default) or the BiLSTM (E5). The app cleans the text with the same functions used in training (`src/preprocess.py`), runs the saved model through `src/inference.py`, and shows the predicted topic in English and Kinyarwanda, a score bar, the top five topics and both models' predictions side by side. For the test examples it also shows whether the prediction matches the dataset label, so both correct and wrong cases can be demonstrated, and it warns when a text is very short. **Results** and **The data** show the experiment results, per-topic F1 and the cleaning statistics in interactive charts. The fine-tuned Transformer (E6) is not in the live app: it was the weakest model and needs more memory than free hosting provides.

9. Limitations and Future Work

Data. After removing duplicates, only 6.7k training articles remain, and six topics have fewer than 200. The test labels follow each website's own idea of the categories, so some of the remaining error is label noise. Each article also gets one topic, even when topics overlap. **Models.** Because we only had a CPU, the Transformers saw just the first 128 subword tokens (about 50 words, against a median of about 300) and had their embeddings frozen. Full fine-tuning on a GPU with 512-token inputs, or splitting long articles into chunks, could change which model comes out on top. **Evaluation.** Each setup was run once with one seed and without a hyperparameter search, so differences smaller than about one macro-F1 point shouldn't be read too much into. The SVM's scores in the app are not calibrated probabilities. **Future work.** The next steps would be: longer-context and morphology-aware models such as KinyaBERT; training on more news sources, or adapting to new sources using their unlabelled text; allowing several topics per article or merging overlapping categories; and a proper learning-rate search (between 5e-5 and 1.5e-4) so that AfroXLMR-mini and mMiniLM can be compared at a setting where both learn.

10. Conclusion

We built and deployed a Kinyarwanda news topic classifier and compared eight experiments on a cleaned version of KINNEWS. The most important lesson was about the data rather than the models: more than half of the training rows were duplicates, and accuracy near the published ~88% only appears when those duplicates leak into the evaluation. The best model was also one of the simplest: a linear SVM on character n-grams, which handles Kinyarwanda's many word forms well and reached 76.1% accuracy and 0.577 macro-F1. Class weights and Word2Vec embeddings both gave clear gains, but neither the BiLSTM nor the small CPU-trained Transformers beat the SVM. Most of the remaining errors come from new websites and from websites using categories differently, not from the models misreading the language. What we take away from this project:

check the data before modelling, test on data that looks like real use (new sources, no duplicates), and don't assume a bigger model will beat a well-tuned simple one when data and compute are limited.

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